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► To cite this version:

Tristan Bullion-Gauthier, Kai Xiao. L^2 -regularity of minimizers of anisotropic interaction functionals. 2026. [⟨hal-05743076⟩](#)

HAL Id: hal-05743076

<https://hal.science/hal-05743076v1>

Preprint submitted on 8 Sep 2026

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L^2 –regularity of minimizers of anisotropic interaction functionals

Tristan Bullion-Gauthier

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September 8, 2026

Abstract

We consider interaction functionals of the form

$$\nu \mapsto \mathcal{E}(\nu) = \int_{\mathbb{R}^N} \int_{\mathbb{R}^N} W(x-y) d\nu(y) d\nu(x) + \int_{\mathbb{R}^N} V(x) d\nu(x),$$

involving an anisotropic kernel W and a general confinement potential V . Under standard assumptions on W and its Fourier transform, we show that the minimizer of $\mathcal{E}$ has L^2 density. Our argument relies on a careful analysis of nonlocal variational inequalities, which may be of independent interest.

1 Introduction

Setting and main result. We study the regularity of minimizers of the functional given by

$$\mathcal{P}(\mathbb{R}^N) \ni \nu \mapsto \mathcal{E}(\nu) := \int_{\mathbb{R}^N} \int_{\mathbb{R}^N} W(x-y) d\nu(y) d\nu(x) + \int_{\mathbb{R}^N} V(x) d\nu(x). \quad (1.1)$$

Here, $N \geq 3$. We assume that

$$V \in (C^0 \cap H_{\text{loc}}^2)(\mathbb{R}^N; \mathbb{R}_+) \text{ is a non-negative coercive function.} \quad (\text{HV})$$

Concerning W , we assume that it is of the form

$$W(x) = \frac{\Psi(x/|x|)}{|x|^{N-2}}, \text{ where } \Psi \in C(\mathbb{S}^{N-1}) \text{ is even and positive.} \quad (\text{HW})$$

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Keywords: Nonlocal energies, anisotropic interactions, variational inequalities

Acknowledgments: We thank Petru Mironescu for introducing us to this research topic and for his interesting questions which are at the core of the present article. We are grateful to him for his careful reading and suggestions to improve the presentation. We also thank Sofiane Cherf for many insightful discussions.

Use of AI: No generative AI was used in the preparation of this article.

We also assume that the Fourier transform of W is of the form

$$\begin{aligned} \mathcal{F}(W)(\xi) &= \frac{m(\xi/|\xi|)}{|\xi|^2}, \text{ where } m \in L^\infty(\mathbb{S}^{N-1}) \text{ satisfies} \\ 0 < m_0 \leq m(\omega) \leq m_1 < \infty, \forall \omega \in \mathbb{S}^{N-1}. \end{aligned} \tag{HFW}$$

The fact that $\mathcal{F}(W)$ is a (-2) -homogeneous tempered distribution is classical. When $N \geq 5$, $\mathcal{F}(W)$ belongs to L^1_{loc} , and, in particular, m in (HFW) is an L^1 function. This is also the case when $N = 3, 4$ and Ψ is sufficiently smooth (see, e.g., Lemoine [14, Theorem 3.2.4]), but need not be the case under the sole assumption (HW). The additional conditions that we impose on m are essentially the same as in, e.g., Carillo *et al.* [7], Mateu *et al.* [16], and Frank *et al.* [10]; see the discussion after the statement of Theorem 1.1.

A typical example of a kernel W satisfying (HW) and (HFW) is

$$W_\alpha(x) := \frac{1}{|x|^{N-2}} + \alpha \frac{x_1^2}{|x|^N}, \text{ where } \alpha \in (-1, N-2),$$

was considered in [7]. It satisfies

$$\mathcal{F}(W_\alpha)(\xi) = C_N \frac{(N-2-\alpha)\xi_1^2 + (N-2+\alpha)\sum_{k=2}^N \xi_k^2}{|\xi|^4} > 0,$$

where $C_N > 0$ is a dimensional constant (see Stein [20, (32), p.73]).

Other examples of kernels satisfying (HW) and (HFW) are “small” perturbations of the Coulomb kernel $1/|x|^{N-2}$, e.g.,

$$W(x) = \frac{1}{|x|^{N-2}} + \varepsilon \frac{\Omega(x/|x|)}{|x|}, \text{ where } \Omega \in C^\infty(\mathbb{S}^{N-1}) \text{ and } \varepsilon > 0 \text{ is sufficiently small.}$$

Under the assumptions (HV), (HW), and (HFW), there exists a unique minimizer of the functional $\mathcal{E}$, and this minimizer is compactly supported. In special cases, this was proved by several authors, see, e.g., Serfaty [19], Mora, Rondi, and Scardia [18], Mateu *et al.* [16]. For the convenience of the reader, we present a full proof of this fact in Appendix A.

Our main result is the following.

Theorem 1.1. *Assume that (HV), (HW), and (HFW) hold. Then, the unique minimizer μ of $\mathcal{E}$ has L^2 density.*

Motivation and previous results. The minimization of energies of the form (1.1), when W is the Coulomb kernel $(-\log(|x|))$ in two dimensions, is a classical problem of electrostatics. But these energies also appear in other contexts. For example, they arise in the study of the mean-field description of Coulomb gases, an important model in statistical mechanics (see [19]). They also play an crucial role in mathematical biology for their connection to aggregation equations (central in biological modelling, see Topaz, Bertozzi, and Lewis [21]). This connection motivated Carillo *et al.* to study the existence of minimizers of (1.1), as well

as their uniqueness and regularity properties in, e.g., Balagué *et al.* [2], Carrillo, Delgadino, and Mellet [5], Carrillo and Shu [8].

The first results on the minimizers of (1.1) were obtained in the isotropic case $W(x) = 1/|x|^{N-2}$, starting with the seminal works of Frostman [11]. Recently, several explicit results were obtained in the case of anisotropic repulsive interactions W , following the important contributions of Mora, Rondi, and Scardia [18] and Carrillo *et al.* [6], where special perturbations of Coulomb interactions were considered in two dimensions. For example, in Mateu *et al.* [16] and Carrillo and Shu [9], it is shown that, for $N = 3$, $V(x) = |x|^2$, and under assumptions on W in the spirit of (HW) and (HFW), the unique minimizer of $\mathcal{E}$ is the normalized characteristic function of an ellipsoid. Moreover, it is proved in [16] and [9] that, if the assumption (HFW) is relaxed to $m \geq 0$, then the minimizing measure μ may be singular and thus have no density.

These results were obtained *via* the following strategy: (i) prove that minimality is equivalent to the Euler–Lagrange equation associated with $\mathcal{E}$; (ii) find a solution of the Euler–Lagrange equation within the class of centered ellipsoids. While the first part of this program is robust and can be completed for large classes of potentials V and W , the second one is specific to the case where V is a quadratic form, typically $V(x) = |x|^2$, and its conclusion need not hold for a general V (see, e.g., [16, (2.18)]).

In a different direction, the case of a general V when $W(x) = 1/|x|^{N-2}$ was studied in [5], where it is shown that minimizing $\mathcal{E}$ amounts to solving an obstacle problem. This is exploited in order to derive regularity results. The starting point in [5] is the Euler–Lagrange equation associated with the minimization of $\mathcal{E}$, which formally reads as

$$\begin{cases} W * \mu + 1/2V \geq \alpha := \mathcal{E}(\mu) - 1/2 \int V d\mu, \\ W * \mu + 1/2V = \alpha \text{ on } \text{supp } \mu, \end{cases} \quad (1.2)$$

if μ is the minimizer of $\mathcal{E}$.

In the case where $W = 1/|x|^{N-2}$, using the above Euler–Lagrange equation, it is proved in [5] that the potential $u = W * \mu$ satisfies the following obstacle problem

$$\begin{cases} u \geq \alpha - 1/2V, \\ -\Delta u \geq 0 \text{ in } \mathbb{R}^N, \\ -\Delta u = N(N-2)|B_1|\mu = 0 \text{ in } \{u > \alpha - 1/2V\} \subset (\text{supp } \mu)^c. \end{cases} \quad (1.3)$$

This problem is associated with a *variational inequality*

$$\begin{aligned} w \in H, w \geq \alpha - 1/2V, \\ \int_{\Omega} \nabla w \cdot \nabla (v - w) dx \geq 0, \forall v \in H, v \geq \alpha - 1/2V, \end{aligned} \quad (1.4)$$

for an appropriate domain Ω and H^1 -type space H .

While it is rather straightforward to obtain (1.2) and to derive from it (1.3), getting (1.4)

form (1.3) requires more work. This program has been completed in [5] when W is the Coulomb potential. One of the crucial steps in their approach consists of establishing the continuity of $W * \mu$; this relies on the fact that the Coulomb kernel is, up to a constant, a fundamental solution of the Laplacian. For another approach in this case, see [19, Proposition 2.22], and also Armstrong, Serfaty, and Zeitouni [1, Lemma 2.3]. Once the validity of (1.4) is established, one can appeal to the regularity theory for obstacle problems and variational inequalities, which has been developed since the 1960's starting with the works of Lewy, Stampacchia, Brezis, Kinderlehrer, and Frehse. As an example of use of this theory, it is proved in [5, Theorem 3.4] that, if V is $W_{\text{loc}}^{2,p}$ with $p > N$, then μ is L^∞ .

Comments on the proof of Theorem 1.1. We rely on the link between the minimization of $\mathcal{E}$ and variational inequalities described above. In order to consider anisotropic kernels of the form (HW), we are led to consider operators and linear or bilinear forms that are defined *via* the Fourier transform. Two main difficulties arise in this process.

We face a first difficulty when trying to define a variational inequality involving the potential $u = W * \mu$. When $W = 1/|x|^{N-2}$, u enjoys properties that need not hold for a general W . More specifically, u is superharmonic, and this allows the authors of [5] to prove that u is everywhere continuous. Superharmonicity is also used in the approximation procedure in [19, Proposition 2.22]. This property is not available under the assumptions of Theorem 1.1. In order to bypass this difficulty, we rely on regularity results established in Section 2, results that may be of independent interest.

The second difficulty arises from the fact that the variational inequality associated with the minimization of $\mathcal{E}$ involves linear and bilinear forms defined in terms of the Fourier transform. We cope with such forms using a penalization approach strongly inspired by Kinderlehrer and Stampacchia [12, Chapter IV, Section 5]. This leads to the study of nonlocal penalized problems different from the ones considered in [12, Chapter IV, Section 5]. This requires a careful adaptation of the approach in [12]. An advantage of our approach when compared to the one in [5] is that it does not rely on continuity and thus holds in any dimension $N \geq 3$. A drawback is that it only leads to L^2 estimates, while one would expect L^p -regularity for some $p > 2$ (depending on N).

Our paper is organized as follows. In Section 2, we gather a few auxiliary results and preliminary regularity results. In Section 3, we prove that the potential $u = W * \mu$ satisfies a variational inequality. Finally, Section 4 is devoted to the proof of a regularity result on the solution of this variational inequality, and of Theorem 1.1.

2 First regularity results

In the following, we denote by μ the minimizer of $\mathcal{E}$ and, with no loss of generality, assume that $\text{supp } \mu \subset B_1$, where, for a non-negative measure ν ,

$$\text{supp } \nu := \{x \in \mathbb{R}^N; \nu(B(x, r)) > 0, \forall r > 0\}.$$

We will only consider spaces of real-valued functions and denote $L^2 = L^2(\mathbb{R}^N; \mathbb{R})$, and similarly for L^p , $\dot{H}^1$, etc. We will denote by C any positive constant depending only on N and W . We will sometimes use the notation “ $A \lesssim B$ ” (respectively “ $A \approx B$ ”) to indicate that $A \leq CB$ (respectively $\frac{1}{C}B \leq A \leq CB$) for such a constant C .

We start with a few standard results (Lemmas 2.1–2.5). The first result is in the spirit of [18, Theorem 3.1]

Lemma 2.1. *μ satisfies*

$$W * \mu + 1/2V \geq \alpha := \mathcal{E}(\mu) - 1/2 \int V d\mu, \text{ Lebesgue a.e.}, \quad (2.1)$$

$$W * \mu + 1/2V = \alpha, \mu - \text{a.e.}, \quad (2.2)$$

$$W * \mu(x) + 1/2V(x) \leq \alpha, \forall x \in \text{supp } \mu. \quad (2.3)$$

Proof. Equations (2.1) and (2.2) follow from the proof of [18, Theorem 3.1]. Equation (2.3) is a consequence of (2.2), thanks to the l.s.c. of $W * \mu + 1/2V$, as shown in [2, Theorem 4]. $\square$

The next result is straightforward and we omit its proof.

Lemma 2.2. *Let ν be a non-negative measure such that*

$$\int_{\mathbb{R}^N} W * \nu d\nu < \infty.$$

Then, $\nu(\{x\}) = 0$ for each $x \in \mathbb{R}^N$.

We next prove a first regularity result, which is a straightforward consequence of Landkof [13, Lemma 1.7]. We set $E(x) := 1/|x|^{N-2}$.

Lemma 2.3. *Let ν be a non-negative and finite measure such that*

$$\int_{\mathbb{R}^N} W * \nu d\nu < \infty.$$

*Then, $W * \nu \in \dot{H}^1 := \{u \in L^{2N/(N-2)}; \nabla u \in L^2\}$.*

Proof. We have

$$\int_{\mathbb{R}^N} |\nabla(E * \nu)|^2 dx \leq C \int_{\mathbb{R}^N} E * \nu d\nu \leq C \int_{\mathbb{R}^N} W * \nu d\nu < \infty, \quad (2.4)$$

where the first inequality follows from [13, Lemma 1.7], and the second inequality is obtained thanks to (HW). The conclusion follows by approximation of ν with the convolutions $\nu_\varepsilon := \nu * \rho_\varepsilon$, where ρ is a standard mollifier. Note that, since $W \in \mathcal{S}'$ and $\nu_\varepsilon \in \mathcal{S}$, we have $\mathcal{F}(W * \nu_\varepsilon) = \mathcal{F}(W)\mathcal{F}(\nu_\varepsilon)$ in the sense of tempered distributions and thus $\mathcal{F}(W * \mu_\varepsilon)$ is a

function. For each $\varepsilon > 0$, we have

$$\begin{aligned}
\frac{1}{(2\pi)^N} \int_{\mathbb{R}^N} |\xi|^2 |\mathcal{F}(W * \nu_\varepsilon)(\xi)|^2 d\xi &= \int_{\mathbb{R}^N} |\xi|^2 |\mathcal{F}(W)(\xi)|^2 |\mathcal{F}(\nu_\varepsilon)(\xi)|^2 d\xi \\
&\leq C \int_{\mathbb{R}^N} |\xi|^2 |\mathcal{F}(E)(\xi)|^2 |\mathcal{F}(\nu_\varepsilon)(\xi)|^2 d\xi \\
&= (2\pi)^N \int_{\mathbb{R}^N} |\nabla(E * \nu_\varepsilon)|^2 dx \\
&\leq (2\pi)^N \int_{\mathbb{R}^N} |\nabla(E * \nu)|^2 dx.
\end{aligned}$$

Here, we rely on the fact that m (defined in (HFW)) is bounded for the first inequality. The last identity follows from Plancherel's theorem, while the last inequality is a standard convolution estimate. By (2.4), we thus find that $\nabla(W * \nu_\varepsilon)$ is bounded in L^2 and this implies that $\nabla(W * \nu)$ is in L^2 , since $W * \nu_\varepsilon \rightarrow W * \nu$ in $\mathcal{S}'$.

In order to show that $W * \nu$ is in $L^{2N/(N-2)}$, we note that $W = W \mathbb{1}_{B_1} + W \mathbb{1}_{(B_1)^c} \in L^1 + L^p$ for any $p > N/(N-2)$. Since ν is a finite measure, we also have $W * \nu \in L^1 + L^p$. Hence, $\{x \in \mathbb{R}^N; W * \nu(x) > t\}$ is of finite Lebesgue measure for each $t > 0$ ($W * \mu$ is “vanishing at infinity”) and the Sobolev embedding (see, e.g., Leoni [15, Theorem 12.4]) yields $W * \nu \in L^{2N/(N-2)}$. $\square$

Lemma 2.4. ([13, Theorem 1.4]) *Let ν be a non-negative measure. Then*

$$E * \nu(x) \geq \int_{B(x,r)} E * \nu(z) dz, \forall x \in \mathbb{R}^N, r > 0.$$

We now turn to an important auxiliary result in the vein of Evans' theorem (see, e.g., Caffarelli [4, Theorem 1]). Its conclusion, combined with Lemma 2.6, leads to a variational inequality that will allow us to complete the proof of Theorem 1.1.

Lemma 2.5. *We have*

$$W * \mu \text{ is continuous in } (\text{supp } \mu)^c. \quad (2.5)$$

$$W * \mu \text{ is continuous at every } x \in \text{supp } \mu \text{ such that } W * \mu(x) + 1/2V(x) = \alpha. \quad (2.6)$$

$$W * \mu \text{ is continuous } \mu - \text{a.e. in } \text{supp } \mu. \quad (2.7)$$

In the proof of Lemma 2.5 (ii), we will use the fact that $E * \mu$ is harmonic (in the classical sense) in $(\text{supp } \mu)^c$ (which is an open set), and therefore

$$E * \mu(x) = \int_{B(x, \text{dist}(x, \text{supp } \mu))} E * \mu(z) dz \quad (2.8)$$

for each $x \in (\text{supp } \mu)^c$.

Proof of Lemma 2.5. Proof of (2.5). For each $x \in (\text{supp } \mu)^c$ and each sequence $x_n \rightarrow x$, there

exists $C_x < \infty$, such that, for n sufficiently large,

$$0 \leq W(x_n - y) \leq C_x < \infty, \forall y \in \text{supp } \mu.$$

This inequality, combined with the fact that W is continuous in $\mathbb{R}^N \setminus \{0\}$, yields the desired conclusion by dominated convergence.

Proof of (2.6). Let $x \in \text{supp } \mu$ be such that $W * \mu(x) + 1/2V(x) = \alpha$. Let $x_n \rightarrow x$. We first observe that, by l.s.c. of $W * \mu$, we have

$$\lim_n W * \mu(y_n) = W * \mu(x), \quad (2.9)$$

for each $(y_n) \subset \text{supp } \mu$ converging to x . Indeed, we have

$$\begin{aligned} \alpha - 1/2V(x) = W * \mu(x) &\leq \liminf_n W * \mu(y_n) \leq \limsup_n W * \mu(y_n) \\ &\leq \lim_n (\alpha - 1/2V(y_n)) = \alpha - 1/2V(x), \end{aligned}$$

using the l.s.c. of $W * \mu$ for the first inequality and (2.3).

Thus, in order to prove (2.6), we only need to consider $x_n \rightarrow x$ with $(x_n) \subset (\text{supp } \mu)^c$. It suffices to check that

$$\forall \varepsilon > 0, \exists \delta > 0, \int_{B(x, \delta)} W(x_n - y) d\mu(y) < \varepsilon, \forall n \text{ sufficiently large.} \quad (2.10)$$

Indeed, granted (2.10), we have that, for each $\varepsilon > 0$, there exists $\delta > 0$ such that

$$\begin{aligned} |W * \mu(x_n) - W * \mu(x)| &\leq \left| \int_{B(x, \delta)^c} W(x_n - y) d\mu(y) - \int_{B(x, \delta)^c} W(x - y) d\mu(y) \right| \\ &\quad + 2\varepsilon, \end{aligned}$$

for n sufficiently large. On the other hand, for n sufficiently large, we have $x_n \in B(x, \delta/2)$ and thus $|x_n - y| \geq \delta/2$, for each $y \in B(x, \delta)^c$. Using the dominated convergence theorem, we therefore have

$$\lim_n \left| \int_{B(x, \delta)^c} W(x_n - y) d\mu(y) - \int_{B(x, \delta)^c} W(x - y) d\mu(y) \right| \rightarrow 0. \quad (2.11)$$

Hence, passing to the limits in (2.11), we find

$$\limsup_n |W * \mu(x_n) - W * \mu(x)| \leq 2\varepsilon, \forall \varepsilon > 0,$$

and this implies (2.6).

Proof of (2.10). Let $\varepsilon > 0$. By dominated convergence, there exists $\delta > 0$ sufficiently small

such that

$$\int_{B(x,\delta)} W(x-y) d\mu(y) < \varepsilon. \quad (2.12)$$

Indeed, by (2.3), we have $W(x-\cdot) \in L^1(\mu)$. On the other hand, $\mathbb{1}_{B(x,\delta)}(y) \rightarrow 0$ for μ -a.e. $y \in \mathbb{R}^N$ since $\mu(\{x\}) = 0$ by Lemma 2.2, which implies (2.12).

For each $n \in \mathbb{N}$, set $r_n := \text{dist}(x_n, \text{supp } \mu)$ and let $y_n \in \text{supp } \mu$ be a point such that $|y_n - x_n| = r_n$. We have

$$\begin{aligned} \int_{B(x,\delta)} W(x_n - y) d\mu(y) &\stackrel{(a)}{\lesssim} \int_{B(x,\delta)} E(x_n - y) d\mu(y) = E * \mu|_{B(x,\delta)}(x_n) \\ &\stackrel{(b)}{=} \int_{B(x_n, r_n)} (E * \mu|_{B(x,\delta)})(z) dz \\ &\leq 2^N \int_{B(y_n, 2r_n)} (E * \mu|_{B(x,\delta)})(z) dz \\ &\stackrel{(c)}{\lesssim} (E * \mu|_{B(x,\delta)})(y_n) \\ &\stackrel{(d)}{\lesssim} \int_{B(x,\delta)} W(y_n - y) d\mu(y) \\ &= W * \mu(y_n) - \int_{B(x,\delta)^c} W(y_n - y) d\mu(y). \end{aligned} \quad (2.13)$$

Here : (a) and (d) follow from (HW); (b) follows from the fact that $E * \mu|_{B(x,\delta)}$ is harmonic and continuous outside of $\text{supp } \mu$; (c) is a consequence of Lemma 2.4. By (2.9), we have

$$\lim_n W * \mu(y_n) = W * \mu(x) = \int_{B(x,\delta)} W(x-y) d\mu(y) + \int_{B(x,\delta)^c} W(x-y) d\mu(y). \quad (2.14)$$

On the other hand, we have

$$\left| \int_{B(x,\delta)^c} W(y_n - y) d\mu(y) - \int_{B(x,\delta)^c} W(x - y) d\mu(y) \right| \rightarrow 0, \text{ as } n \rightarrow \infty. \quad (2.15)$$

by dominated convergence, as in (2.11).

Combining (2.13), (2.14) and (2.15), we find

$$\begin{aligned} \int_{B(x,\delta)} W(x_n - y) d\mu(y) &\lesssim W * \mu(x) - \int_{B(x,\delta)^c} W(y_n - y) d\mu(y) + o_{n \rightarrow \infty}(1) \\ &= \int_{B(x,\delta)} W(x - y) d\mu(y) + o_{n \rightarrow \infty}(1), \end{aligned}$$

which yields (2.10), by (2.12). □

Similar arguments lead to the following result.

Lemma 2.6. $W * \mu$ is bounded (everywhere).

Proof. For each $x \in \text{supp } \mu \subset B_1$, by (2.7), we have

$$0 \leq (W * \mu)(x) \leq \alpha - \frac{1}{2}V(x) \leq \alpha + \frac{1}{2}\|V\|_{L^\infty(B_1)}.$$

For $x \in (\text{supp } \mu)^c$, we denote $r := \text{dist}(x, \text{supp } \mu)$ and consider $y \in \text{supp } \mu$ such that $|x - y| = \text{dist}(x, \text{supp } \mu)$.

We have

$$\begin{aligned} (W * \mu)(x) &\lesssim (E * \mu)(x) \leq \int_{B(x,r)} (E * \mu)(z) \, dz \leq 2^N \int_{B(y,2r)} (E * \mu)(z) \, dz \\ &\stackrel{(a)}{\leq} 2^N (E * \mu)(y) \lesssim W * \mu(y) \leq \alpha - \frac{1}{2}V(y) \leq \alpha + \frac{1}{2}\|V\|_{L^\infty(B_1)}. \end{aligned}$$

Here, we rely on Lemma 2.4 for (a). □

3 A variational inequality

This section is strongly inspired by [5]. Its purpose is to point out a variational problem, similar to (1.4), that will allow us to show that μ has an L^2 density.

We start with an heuristic derivation of a variational inequality involving $W * \mu$ when $W(x) = 1/|x|^{N-2}$. In this case, W is such that $-\Delta W = N(N-2)|B_1|\delta_0$. This fact, combined with the Euler–Lagrange equation (2.1), implies that $w = W * \mu$ solves the obstacle problem (1.3). Assuming for simplicity that w is C_c^∞ , we find that, for each $v \in \dot{H}^1$ such that $v \geq \alpha - 1/2V$,

$$\int_{\mathbb{R}^N} \nabla w \cdot \nabla (v - w) \, dx = \int_{\mathbb{R}^N} (-\Delta W)(v - w) \, dx \geq 0, \quad (3.1)$$

since $-\Delta w$ is non-negative and supported in $\text{supp } \mu \subset \{w = \alpha - 1/2V\}$, where $v \geq w$.

In this section, we introduce a bilinear form a (see (3.4)) in relation with W . More precisely, a and W will be related in a way that mimics the connection between the Dirichlet form

$$(v, w) \mapsto \int_{\mathbb{R}^N} \nabla v \cdot \nabla w \, dx \quad (3.2)$$

and the fundamental solution of $-\Delta$ (see (3.7) and (3.9)).

Next, we justify an integration by parts in the vein of (3.1), which allows us to show that, when W satisfies (HW) and (HFW), $W * \mu$ solves a variational inequality similar to the above, where the Dirichlet form (3.2) is replaced with a . For this purpose, we rely on Lemmas 2.3–2.6, and our approach is slightly different from the ones that have been used to deal with the case $W = 1/|x|^{N-2}$ (see, e.g., [19, Proposition 2.22]). The main issue that forces us to take a different route is the fact that $W * \mu$ is not superharmonic.

Recall that we have assumed that $\text{supp } \mu \subset B_1$. Let u_0 be a smooth function such that

$$0 \leq u_0 \leq \alpha, u_0 = \alpha \text{ in } \text{supp } \mu, \text{ and } u_0 = 0 \text{ in } B_2^c.$$

We also set $f = 1/2\eta V$, where η is a cut-off function such that $\eta = 1$ in B_4 . Since $V \in H_{\text{loc}}^2$, we have $f \in H^2$. By (2.1) and the fact that W is positive, we have

$$\begin{aligned} W * \mu + f &\geq u_0, \text{ Lebesgue a.e.}, \\ W * \mu + f &= u_0, \mu - \text{a.e. in } \text{supp } \mu. \end{aligned}$$

We set

$$u := W * \mu + f - u_0. \quad (3.3)$$

We will show that u solves a variational inequality involving the symmetric bilinear form

$$a: \dot{H}^1 \times \dot{H}^1 \ni (v, w) \mapsto \int_{\mathbb{R}^N} \frac{|\xi|^2}{m(\xi/|\xi|)} \mathcal{F}(v)(\xi) \overline{\mathcal{F}(w)(\xi)} d\xi. \quad (3.4)$$

We also set

$$F(v) := \int_{\mathbb{R}^N} \frac{|\xi|^2}{m(\xi/|\xi|)} \mathcal{F}(f - u_0)(\xi) \overline{\mathcal{F}(v)(\xi)} d\xi = a(f - u_0, v). \quad (3.5)$$

The bilinear form a is associated with the linear operator A defined, at least for $v \in L^2$ or $v \in \dot{H}^1$, by

$$\mathcal{F}(Av)(\xi) = (2\pi)^N \frac{|\xi|^2}{m(\xi/|\xi|)} \mathcal{F}(v)(\xi). \quad (3.6)$$

We have, by Plancherel's formula,

$$a(v, w) = \int_{\mathbb{R}^N} (Av)w dx, \quad (3.7)$$

for each sufficiently smooth $v \in \dot{H}^2$ and $w \in L^2$, where

$$\dot{H}^2 := \{v \in L_{\text{loc}}^1; \partial_{ij}v \in L^2, \forall 1 \leq i, j \leq N\}. \quad (3.8)$$

The connection between a and W is provided by the formula

$$A\left(\frac{W}{(2\pi)^N}\right) = \delta_0, \quad (3.9)$$

which will play a role in what follows.

We first gather some obvious properties of a and F . These properties will be instrumental

in the following.

Lemma 3.1. *a is symmetric, real-valued and satisfies*

$$a(v, v) \approx \|\nabla v\|_{L^2}^2, \forall v \in \dot{H}^1. \quad (3.10)$$

F is also real-valued. Moreover,

$$F \text{ is linear and continuous on } L^2 \text{ and } \dot{H}^1. \quad (3.11)$$

(We will denote by $\|F\|_{(L^2)^*}$ and $\|F\|_{(\dot{H}^1)^*}$ its corresponding norms.)

Proof. a is indeed symmetric and real-valued since

$$\begin{aligned} \overline{a(v, w)} &= a(w, v) = \int_{\mathbb{R}^N} \frac{|\xi|^2}{m(\xi/|\xi|)} \mathcal{F}(w)(\xi) \overline{\mathcal{F}(v)(\xi)} d\xi \\ &\stackrel{(a)}{=} \int_{\mathbb{R}^N} \frac{|-\xi|^2}{m(-\xi/|-\xi|)} \overline{\mathcal{F}(w)(-\xi)} \mathcal{F}(v)(-\xi) d\xi \\ &= a(v, w). \end{aligned} \quad (3.12)$$

For (a), we rely on the fact that $\mathcal{F}(w)(-\xi) = \overline{\mathcal{F}(w)(\xi)}$, for any real-valued function $w \in \dot{H}^1$, and on the evenness of m . The same computation shows that F is real-valued.

The estimate

$$a(v, v) \approx \|\nabla v\|_{L^2}^2, \forall v \in \dot{H}^1,$$

is a straightforward consequence of (HFW).

Using (HFW) and the Cauchy–Schwarz inequality, we find that F is continuous on L^2 and $\dot{H}^1$, since $f - u_0$ is in H^2 . $\square$

We now state the main result of this section, inspired by [5, Proposition 3.3].

Lemma 3.2. *u is the unique solution of the variational inequality*

$$u \geq 0, a(u, v - u) \geq F(v - u), \forall v \in \dot{H}^1, v \geq 0 \text{ a.e.} \quad (3.13)$$

In order to prove Lemma 3.2, we establish the following auxiliary result.

Lemma 3.3. *For each $\varphi \in C_c^\infty$, we have*

$$a(W * \mu, \varphi) = (2\pi)^N \int_{\mathbb{R}^N} \varphi d\mu.$$

Proof. Set $\mu_\varepsilon = \mu * \rho_\varepsilon$. By Lemma 2.3, $W * \mu \in \dot{H}^1$, thus $W * \mu_\varepsilon \in \dot{H}^1$ and $W * \mu_\varepsilon \rightarrow W * \mu$ in $\dot{H}^1$. Therefore, since a is continuous on $\dot{H}^1$, we find that

$$a(W * \mu_\varepsilon, \varphi) \rightarrow a(W * \mu, \varphi), \text{ as } \varepsilon \rightarrow 0. \quad (3.14)$$

On the other hand, we clearly have

$$\int_{\mathbb{R}^N} \varphi(x) \mu_\varepsilon(x) dx \rightarrow \int_{\mathbb{R}^N} \varphi d\mu, \quad (3.15)$$

since $\varphi \in C_c^\infty$.

Finally,

$$\begin{aligned} a(W * \mu_\varepsilon, \varphi) &= \int_{\mathbb{R}^N} \frac{|\xi|^2}{m(\xi/|\xi|)} \mathcal{F}(W * \mu_\varepsilon)(\xi) \overline{\mathcal{F}(\varphi)(\xi)} d\xi \\ &= \int_{\mathbb{R}^N} \frac{|\xi|^2}{m(\xi/|\xi|)} \mathcal{F}(W)(\xi) \mathcal{F}(\mu_\varepsilon)(\xi) \overline{\mathcal{F}(\varphi)(\xi)} d\xi \\ &= \int_{\mathbb{R}^N} \mathcal{F}(\mu_\varepsilon)(\xi) \overline{\mathcal{F}(\varphi)(\xi)} d\xi = (2\pi)^N \int_{\mathbb{R}^N} \varphi(x) \mu_\varepsilon(x) dx. \end{aligned} \quad (3.16)$$

Passing to the limits in (3.16) yields the desired conclusion, using (3.14) and (3.15). $\square$

Proof of Lemma 3.2. It suffices to show that (3.13) holds for each nonnegative $v \in C_c^\infty$. Indeed, this implies the desired conclusion thanks to: (i) the continuity on $\dot{H}^1$ of the forms a and F ; (ii) the fact that, for each nonnegative $v \in \dot{H}^1$, we may find a sequence $(v_n) \subset C_c^\infty$ of nonnegative functions such that $v_n \rightarrow v$ in $\dot{H}^1$.

Let $v \in C_c^\infty$ be such that $v \geq 0$. We have

$$\begin{aligned} a(u, v) &= a(W * \mu + f - u_0, v) = a(W * \mu, v) + a(f - u_0, v) \\ &= a(W * \mu, v) + F(v). \end{aligned} \quad (3.17)$$

By Lemma 3.3, we have $a(W * \mu, v) = (2\pi)^N \int_{\mathbb{R}^N} v d\mu$ and thus, since $v \geq 0$,

$$a(u, v) \geq F(v). \quad (3.18)$$

Next, we show that

$$a(u, u) = F(u). \quad (3.19)$$

For this purpose, we consider $u_n = \theta(\cdot/n)(u * \rho_{1/n})$, $n \geq 1$, where θ is a cut-off function such that $\theta = 1$ in B_1 . The sequence $(u_n) \subset C_c^\infty$ is such that $u_n \geq 0$, for each $n \geq 1$, and $u_n \rightarrow u$ in $\dot{H}^1$ as $n \rightarrow \infty$ (see, e.g., [15, proof of Theorem 11.43]). By (3.17) and Lemma 3.3, we have, for each $n \geq 1$,

$$a(u, u_n) = a(W * \mu, u_n) + F(u_n) = (2\pi)^N \int_{\mathbb{R}^N} u_n d\mu + F(u_n). \quad (3.20)$$

By Lemma 2.5, we have that $u_n(x) \rightarrow u(x)$ for μ -a.e. $x \in \mathbb{R}^N$ (since u is continuous μ -a.e. and (u_n) is defined by mollifying u). Since u is bounded (by Lemma 2.6 and the definition of u), we have the existence of $C < \infty$ such that $|u_n(x)| \leq C$ for each $x \in \mathbb{R}^N$, $n \geq 1$. Hence,

we have, by dominated convergence, that

$$\int_{\mathbb{R}^N} u_n \, d\mu \rightarrow \int_{\mathbb{R}^N} u \, d\mu, \text{ as } n \rightarrow \infty.$$

But $u = 0$, μ -a.e., by definition of u and Lemma 2.1. Thus

$$\lim_n \int_{\mathbb{R}^N} u_n \, d\mu = 0. \quad (3.21)$$

Since $u_n \rightarrow u$ in $\dot{H}^1$, we also have

$$a(u, u_n) \rightarrow a(u, u) \text{ and } F(u_n) \rightarrow F(u), \text{ as } n \rightarrow \infty. \quad (3.22)$$

Passing to the limits $n \rightarrow \infty$ in (3.20), using (3.21) and (3.22), we find (3.19).

Combining (3.18) and (3.19), we find

$$a(u, v - u) = a(u, v) - a(u, u) \geq F(v) - F(u) = F(v - u).$$

Finally, the uniqueness of u is standard. Indeed, for any other solution $w \in \dot{H}^1$ of (3.13), using

$$\begin{aligned} a(u, w - u) &\geq F(w - u), \\ a(w, u - w) &\geq F(u - w), \end{aligned}$$

we obtain that $a(u - w, u - w) \leq 0$. Using (3.10), we find that $u = w$ a.e. $\square$

4 Regularity of the solution of the variational inequality (3.13)

The main result of this section is the following.

Lemma 4.1. *The solution u of (3.13) is in $\dot{H}^1 \cap \dot{H}^2$ (see (3.8) for the definition of $\dot{H}^2$).*

Remark 4.2. Let us emphasize that the proof of Lemma 4.1 only relies on the fact that u solves (3.13), and not on its relation with our initial minimization problem. More specifically, the proof shows that, if w solves

$$w \in \dot{H}^1, w \geq 0 \text{ a.e.}, a(w, v - w) \geq G(v - w), \forall v \in \dot{H}^1, v \geq 0 \text{ a.e.},$$

with $G \in (\dot{H}^1)^* \cap (L^2)^*$, then $w \in \dot{H}^1 \cap \dot{H}^2$. $\square$

While Lemma 4.1 is interesting on its own, its relevance to our study arises in connection with Theorem 1.1.

Proof of Theorem 1.1 granted Lemma 4.1. We have that $u \in \dot{H}^2$, thanks to Lemma 4.1. We thus

have, for each $v \in L^2$,

$$|a(u, v)| = \left| \int_{\mathbb{R}^N} \frac{|\xi|^2}{m(\xi/|\xi|)} \mathcal{F}(u)(\xi) \overline{\mathcal{F}(v)(\xi)} d\xi \right| \leq C \|D^2 u\|_{L^2} \|v\|_{L^2}, \quad (4.1)$$

thanks to the Cauchy–Schwarz inequality and assumption (HFW). By the definitions (3.3) of u and (3.5) of F , and thanks to Lemma 3.3, we have, for each $\varphi \in C_c^\infty$,

$$a(u, \varphi) = a(W * \mu, \varphi) + F(\varphi) = (2\pi)^N \int_{\mathbb{R}^N} \varphi d\mu + F(\varphi). \quad (4.2)$$

Combining (4.1) with (4.2), and using the continuity of F on L^2 , we finally find that

$$\left| \int_{\mathbb{R}^N} \varphi d\mu \right| \leq C \|\varphi\|_{L^2}, \quad \forall \varphi \in C_c^\infty.$$

This shows that $\mu \in L^2$. □

In order to prove Lemma 4.1, we use an approach that is strongly inspired by [12, Chapter IV, Section 5]. It consists of letting $\varepsilon \rightarrow 0$ in the penalized variational problem (4.4) below. Since u is the solution of (3.13), it is the unique minimizer of

$$E: \dot{H}^1 \ni v \mapsto \frac{1}{2} a(v, v) - F(v),$$

in $\{v \in \dot{H}^1; v \geq 0\}$. We will “approximate” the variational problem

$$\text{minimize } E(v) \text{ on } \{v \in \dot{H}^1; v \geq 0\}, \quad (4.3)$$

by the sequence of problems

$$\text{minimize } E_\varepsilon(v) := \frac{1}{2} a(v, v) + \frac{1}{2} \varepsilon \int_{\mathbb{R}^N} (Tv)v dx + \frac{1}{2\varepsilon} \int_{\{v < 0\}} v^2 dx - F(v) \text{ on } H^1. \quad (4.4)$$

In the above, the term $\frac{1}{2\varepsilon} \int_{\{v < 0\}} v^2 dx$ leads, as $\varepsilon \rightarrow 0$, to the constraint $v \geq 0$ in (4.3). Compared to (4.3), the advantage of (4.4) is that it is straightforward to obtain regularity results on its solution. Our aim is to transfer these results to u via the following strategy.

Step 1. We obtain $\dot{H}^2$ ε -independent estimates on the solution u_ε of (4.4).

Step 2. We extract, as $\varepsilon \rightarrow 0$, a subsequence of (u_ε) that converges to $\tilde{u}$ (in some weak sense), and we show that $\tilde{u}$ is a solution of (3.13). Since u is the unique solution of (3.13), we obtain that $u = \tilde{u}$. The uniform $\dot{H}^2$ estimates on (u_ε) obtained in the first step, and the fact that u_ε converges to u yield $u \in \dot{H}^2$.

We start with Step 1 and first define T by

$$\mathcal{F}(T(v))(\xi) := \frac{1}{m(\xi/|\xi|)} \mathcal{F}(v)(\xi), \text{ for a.e. } \xi \in \mathbb{R}^N \setminus \{0\},$$

for each $v \in L^2$. The following result is the consequence of computations similar to (3.12). We omit its proof.

Lemma 4.3. (i) For each $u \in L^2$, Tu is real valued and $\|Tu\|_{L^2} \approx \|u\|_{L^2}$.

(ii) For each $u \in H^1$, $\|Tu\|_{H^1} \approx \|u\|_{H^1}$.

(iii) The bilinear form

$$L^2 \times L^2 \ni (u, v) \mapsto \int_{\mathbb{R}^N} (Tu)v \, dx$$

is real-valued and symmetric, and we have

$$\int_{\mathbb{R}^N} (Tu)u \, dx \approx \|u\|_{L^2}^2, \forall u \in L^2. \quad (4.5)$$

The same holds for T^{-1} .

The next result yields the approximating sequence (u_ε) .

Lemma 4.4. Let $\varepsilon > 0$. There exists a unique minimizer $u_\varepsilon \in H^1$ of

$$E_\varepsilon: H^1 \ni v \mapsto \frac{1}{2}a(v, v) + \frac{1}{2}\varepsilon \int_{\mathbb{R}^N} (Tv)v \, dx + \frac{1}{2\varepsilon} \int_{\{v < 0\}} v^2 \, dx - F(v). \quad (4.6)$$

Moreover, u_ε is the unique solution of the corresponding Euler–Lagrange equation

$$a(u_\varepsilon, v) + \varepsilon \int_{\mathbb{R}^N} (Tu_\varepsilon)v \, dx - \frac{1}{\varepsilon} \int_{\mathbb{R}^N} u_\varepsilon^- v \, dx = F(v), \forall v \in H^1, \quad (4.7)$$

Formally, equation (4.7) reads

$$Au_\varepsilon + \varepsilon Tu_\varepsilon - \frac{1}{\varepsilon} u_\varepsilon^- = F,$$

with A defined in (3.6).

Proof of Lemma 4.4. Existence of u_ε . By (3.10), (4.5), and since F is continuous on $\dot{H}^1$, there exist constants $0 < c_1, c_2 < \infty$ such that

$$c_1 \|\nabla v\|_{L^2}^2 + c_1 \|v\|_{L^2}^2 - c_2 \|\nabla v\|_{L^2} \leq E_\varepsilon(v), \forall v \in H^1. \quad (4.8)$$

Consider now a minimizing sequence $(v_n) \subset H^1$ of E_ε . This sequence is bounded in H^1 . Indeed, (4.8) implies that $(\|\nabla v_n\|_{L^2})$ is bounded. Since $(\|\nabla v_n\|_{L^2})$ is bounded, we also obtain that $(\|v_n\|_{L^2})$ is bounded, again by (4.8).

Hence, we may extract a subsequence of (v_n) and find $u_\varepsilon \in H^1$ such that $v_n \rightharpoonup u_\varepsilon$ in H^1 . Since every term in E_ε is weakly l.s.c. with respect to H^1 , we find that u_ε minimizes E_ε .

u_ε is unique. This follows from the fact that every term appearing in the definition of E_ε is convex on H^1 , while $v \rightarrow a(v, v)$ (for example) is strictly convex.

u_ε satisfies (4.7). Let $v \in H^1$. We have

$$\begin{aligned} \frac{d}{dt} \left(\frac{1}{2} a(u_\varepsilon + tv, u_\varepsilon + tv) \right) \Big|_{t=0} &= a(u_\varepsilon, v), \\ \frac{d}{dt} \left(\frac{1}{2} \int_{\mathbb{R}^N} (T(u_\varepsilon + tv)) (u_\varepsilon + tv) dx \right) \Big|_{t=0} &= \int_{\mathbb{R}^N} (Tu_\varepsilon) v dx, \\ \frac{d}{dt} (F(u_\varepsilon + tv)) \Big|_{t=0} &= F(v). \end{aligned} \quad (4.9)$$

We also have

$$\begin{aligned} \int_{\mathbb{R}^N} ((u_\varepsilon + tv)^-)^2 dx - \int_{\mathbb{R}^N} (u_\varepsilon^-)^2 dx &= \int_{\{u_\varepsilon < -tv\}} (u_\varepsilon + tv)^2 dx - \int_{\{u_\varepsilon < 0\}} u_\varepsilon^2 dx \\ &= \int_{\{u_\varepsilon < -tv\}} u_\varepsilon^2 dx - \int_{\{u_\varepsilon < 0\}} u_\varepsilon^2 dx + 2t \int_{\{u_\varepsilon < -tv\}} u_\varepsilon v dx + o_{t \rightarrow 0}(t) \\ &= \int_{\{0 \leq u_\varepsilon < -tv\}} u_\varepsilon^2 dx - \int_{\{-tv \leq u_\varepsilon < 0\}} u_\varepsilon^2 dx + 2t \int_{\{u_\varepsilon < -tv\}} u_\varepsilon v dx + o_{t \rightarrow \infty}(t). \end{aligned}$$

But

$$\left| \int_{\{0 \leq u_\varepsilon < -tv\}} u_\varepsilon^2 dx - \int_{\{-tv \leq u_\varepsilon < 0\}} u_\varepsilon^2 dx \right| \leq 2t^2 \int_{\mathbb{R}^N} v^2 dx = o_{t \rightarrow \infty}(t).$$

Therefore, we have

$$\frac{\int_{\mathbb{R}^N} ((u_\varepsilon + tv)^-)^2 dx - \int_{\mathbb{R}^N} (u_\varepsilon^-)^2 dx}{t} = 2 \int_{\{u_\varepsilon < -tv\}} u_\varepsilon v dx + o_{t \rightarrow 0}(1),$$

and we may conclude, by dominated convergence, that

$$\frac{d}{dt} \left(\int_{\mathbb{R}^N} ((u_\varepsilon + tv)^-)^2 dx \right) \Big|_{t=0} = 2 \int_{\{u_\varepsilon < 0\}} u_\varepsilon v dx = -2 \int_{\mathbb{R}^N} u_\varepsilon^- v dx. \quad (4.10)$$

By (4.9) and (4.10), $t \rightarrow E_\varepsilon(u_\varepsilon + tv)$ is differentiable at 0. Since it attains its minimum at 0, by definition of u_ε , (4.9) and (4.10) also yield (4.7). $\square$

We now show ε -independent $\dot{H}^2$ estimates on the solution u_ε of (4.7).

Lemma 4.5. *We have*

$$\|D^2 u_\varepsilon\|_{L^2} \leq C \|F\|_{(L^2)^*}, \quad \forall \varepsilon > 0,$$

where $C < +\infty$ is independent of ε .

Proof. We claim that it suffices to prove that, for each $\varepsilon > 0$, we have

$$\left\| \frac{u_\varepsilon^-}{\varepsilon} \right\|_{L^2} \leq C \|F\|_{(L^2)^*}, \quad (4.11)$$

and

$$\|\varepsilon T u_\varepsilon\|_{L^2} \leq C \|F\|_{(L^2)^*}. \quad (4.12)$$

Indeed, granted (4.11) and (4.12), we have, by (4.7),

$$|a(u_\varepsilon, v)| \leq C \|F\|_{(L^2)^*} \|v\|_{L^2}, \forall v \in H^1. \quad (4.13)$$

Therefore, we have, for each $v \in H^1$,

$$\begin{aligned} \int_{\mathbb{R}^N} \nabla u_\varepsilon \cdot \nabla v \, dx &= \frac{1}{(2\pi)^N} \int_{\mathbb{R}^N} |\xi|^2 \mathcal{F}(u_\varepsilon)(\xi) \overline{\mathcal{F}(v)(\xi)} \, d\xi \\ &= \frac{1}{(2\pi)^N} \int_{\mathbb{R}^N} \frac{|\xi|^2}{m(\xi/|\xi|)} \mathcal{F}(u_\varepsilon)(\xi) \overline{m(\xi/|\xi|) \mathcal{F}(v)(\xi)} \, d\xi \\ &= \frac{1}{(2\pi)^N} \int_{\mathbb{R}^N} \frac{|\xi|^2}{m(\xi/|\xi|)} \mathcal{F}(u_\varepsilon)(\xi) \overline{\mathcal{F}(T^{-1}(v))(\xi)} \, d\xi \\ &= \frac{1}{(2\pi)^N} a(u_\varepsilon, T^{-1}(v)). \end{aligned} \quad (4.14)$$

Hence, we find that

$$\left| \int_{\mathbb{R}^N} \nabla u_\varepsilon \cdot \nabla v \, dx \right| \leq C \|F\|_{(L^2)^*} \|T^{-1}(v)\|_{L^2} \leq C \|F\|_{(L^2)^*} \|v\|_{L^2}, \forall v \in H^1, \quad (4.15)$$

using (4.14) and the fact that T^{-1} is a continuous operator on L^2 (Lemma 4.3). This yields the desired conclusion.

We now prove (4.11) and (4.12). In order to obtain (4.11), we test equation (4.7) with

$$v_1 := -T^{-1} \left(\frac{u_\varepsilon^-}{\varepsilon} \right).$$

The function v_1 is in H^1 since $u_\varepsilon^- \in H^1$ and T^{-1} is a continuous operator on H^1 , by Lemma 4.3. For later use, we recall that

$$\nabla(v^-) = (-\nabla v) \mathbb{1}_{\{v < 0\}}, \forall v \in H^1. \quad (4.16)$$

We have

$$\begin{aligned} a(u_\varepsilon, v_1) &= -\frac{1}{\varepsilon} \int_{\mathbb{R}^N} \frac{|\xi|^2}{m(\xi/|\xi|)} \mathcal{F}(u_\varepsilon)(\xi) \overline{m(\xi/|\xi|) \mathcal{F}(u_\varepsilon^-)(\xi)} \, d\xi \\ &= -\frac{1}{\varepsilon} \int_{\mathbb{R}^N} |\xi|^2 \mathcal{F}(u_\varepsilon)(\xi) \overline{\mathcal{F}(u_\varepsilon^-)(\xi)} \, d\xi \\ &\stackrel{(a)}{=} -\frac{(2\pi)^N}{\varepsilon} \int_{\mathbb{R}^N} \nabla u_\varepsilon \cdot \nabla(u_\varepsilon^-) \, dx \\ &\stackrel{(b)}{=} \frac{(2\pi)^N}{\varepsilon} \int_{\{u_\varepsilon < 0\}} |\nabla u_\varepsilon|^2 \, dx \geq 0. \end{aligned} \quad (4.17)$$

Here, we use Plancherel's formula for (a) and the formula (4.16) for (b).

We also have

$$\begin{aligned} \int_{\mathbb{R}^N} T(u_\varepsilon) v_1 \, dx &= -\frac{1}{\varepsilon(2\pi)^N} \int_{\mathbb{R}^N} \frac{1}{m(\xi/|\xi|)} \mathcal{F}(u_\varepsilon)(\xi) \overline{m(\xi/|\xi|) \mathcal{F}(u_\varepsilon^-)(\xi)} \, d\xi \\ &= -\frac{1}{\varepsilon} \int_{\mathbb{R}^N} u_\varepsilon u_\varepsilon^- \, dx \geq 0. \end{aligned} \quad (4.18)$$

By (4.17), (4.18), and thanks to the equation (4.7), we find that

$$\begin{aligned} \int_{\mathbb{R}^N} \left(-\frac{u_\varepsilon^-}{\varepsilon}\right) T^{-1} \left(-\frac{u_\varepsilon^-}{\varepsilon}\right) \, dx &= \int_{\mathbb{R}^N} \left(-\frac{u_\varepsilon^-}{\varepsilon}\right) v_1 \\ &\leq F(v_1) \\ &\leq \|F\|_{(L^2)^*} \|v_1\|_{L^2} \\ &\leq C \|F\|_{(L^2)^*} \left\| \frac{u_\varepsilon^-}{\varepsilon} \right\|_{L^2}. \end{aligned} \quad (4.19)$$

Combined with the fact that $\int_{\mathbb{R}^N} (T^{-1}v)v \, dx \approx \|v\|_{L^2}^2$, for each $v \in L^2$ (Lemma 4.3), (4.19) yields (4.11).

In order to obtain (4.12), we test equation (4.7) with

$$v_2 := \varepsilon u_\varepsilon \in H^1.$$

We have

$$a(u_\varepsilon, v_2) = \varepsilon a(u_\varepsilon, u_\varepsilon) \geq 0, \quad (4.20)$$

and

$$-\frac{1}{\varepsilon} \int_{\mathbb{R}^N} u_\varepsilon^- v_2 = - \int_{\mathbb{R}^N} u_\varepsilon^- u_\varepsilon \, dx \geq 0. \quad (4.21)$$

Therefore, using (4.7), (4.20), and (4.21), we find that

$$\begin{aligned} \varepsilon^2 \int_{\mathbb{R}^N} (Tu_\varepsilon) u_\varepsilon \, dx &= \varepsilon \int_{\mathbb{R}^N} (Tu_\varepsilon) v_2 \, dx \\ &\leq F(v_2) \leq \|F\|_{(L^2)^*} \|v_2\|_{L^2} = \|F\|_{(L^2)^*} \|\varepsilon u_\varepsilon\|_{L^2}. \end{aligned} \quad (4.22)$$

Since we have that

$$\varepsilon^2 \int_{\mathbb{R}^N} (Tu_\varepsilon) u_\varepsilon \, dx \approx \|\varepsilon u_\varepsilon\|_{L^2}^2 \approx \|\varepsilon Tu_\varepsilon\|_{L^2}^2,$$

by Lemma 4.3, (4.22) implies that (4.12) holds. $\square$

We extract a converging subsequence from $(u_\varepsilon)_{\varepsilon>0}$ thanks to the following result.

Lemma 4.6. *We have*

$$\|\nabla u_\varepsilon\|_{L^2} \leq C\|F\|_{(\dot{H}^1)^*}, \forall \varepsilon > 0,$$

where $C < \infty$ is independent of ε .

Proof. We obtain the desired conclusion by testing (4.7) with $v = u_\varepsilon$, using the same arguments as in the proof of Lemma 4.5. Indeed, since $\int_{\mathbb{R}^N} (Tu_\varepsilon)u_\varepsilon \, dx \geq 0$ (by Lemma 4.3) and $-\int_{\mathbb{R}^N} u_\varepsilon^- u_\varepsilon \, dx \geq 0$, we find, using equation (4.7), that

$$a(u_\varepsilon, u_\varepsilon) \leq F(u_\varepsilon). \quad (4.23)$$

Since $a(u_\varepsilon, u_\varepsilon) \approx \|\nabla u_\varepsilon\|_{L^2}^2$ (by (3.10)) and $F(u_\varepsilon) \leq \|F\|_{(\dot{H}^1)^*} \|\nabla u_\varepsilon\|_{L^2}$, equation (4.23) yields Lemma 4.6. $\square$

We now turn to Step 2. By Lemma 4.6, we find a function $\tilde{u} \in \dot{H}^1$ such that, up to a subsequence,

$$u_{\varepsilon_k} \rightharpoonup \tilde{u} \text{ in } \dot{H}^1. \quad (4.24)$$

We have

$$\|D^2 \tilde{u}\|_{L^2} \leq \liminf_{k \rightarrow \infty} \|D^2 u_{\varepsilon_k}\|_{L^2} \leq C\|F\|_{(L^2)^*}, \quad (4.25)$$

using Lemma 4.5 for the last inequality.

The final ingredient of the proof of Lemma 4.1 is the following.

Lemma 4.7. *The function $\tilde{u}$ satisfies (3.13).*

Proof of Lemma 4.1 granted Lemma 4.7. Since u is the unique solution of (3.13) (Lemma 3.2), Lemma 4.7 implies that $u = \tilde{u}$ a.e. By (4.24) and (4.25), we thus have that $u \in \dot{H}^1 \cap \dot{H}^2$. $\square$

We now turn to the proof of Lemma 4.7. We use the following properties. Recall that, in Hilbert space H , an operator $G : H \rightarrow (H)^*$ is monotone if

$$\langle G(v) - G(w), v - w \rangle \geq 0, \forall v, w \in H, \quad (4.26)$$

and strictly monotone if, in addition, equality in (4.26) implies that $v = w$. We have

Lemma 4.8. *The operator*

$$H^1 \ni v \mapsto G_\varepsilon(v) \in (H^1)^*,$$

defined by

$$\langle G_\varepsilon(v), \varphi \rangle = a(v, \varphi) + \varepsilon \int_{\mathbb{R}^N} (Tv)\varphi - \frac{1}{\varepsilon} \int_{\mathbb{R}^N} v^- \varphi - F(\varphi), \forall \varphi \in H^1,$$

is strictly monotone.

Proof. We notice that

$$a(v - \varphi, v - \varphi) \geq C \int_{\mathbb{R}^N} |\nabla(v - \varphi)|^2 \geq 0, \quad (4.27)$$

and

$$\int_{\mathbb{R}^N} (T(v - \varphi))(v - \varphi) \geq C \int_{\mathbb{R}^N} |v - \varphi|^2 \geq 0. \quad (4.28)$$

Clearly, equality in (4.27) or (4.28) only occurs when $v = \varphi$. Thus, it suffices to prove that

$$(-v^- + \varphi^-)(v - \varphi)(x) \geq 0, \forall x \in \mathbb{R}^N.$$

This is a direct consequence of the fact that $t \mapsto -t^-$ is increasing. $\square$

We now turn to the

Proof of Lemma 4.7. We first prove that $\tilde{u} \geq 0$ a.e. We argue by contradiction and assume that $|\{u < 0\}| > 0$. This implies that there exists $\delta > 0$ such that $|\{\tilde{u} < -\delta\}| > 0$. By (4.24), we have that, up to a subsequence, $u_{\varepsilon_k} \rightarrow \tilde{u}$ a.e. on $A := |\{\tilde{u} < -\delta\}|$, by compactness of the embedding $\dot{H}^1 \hookrightarrow L^2_{\text{loc}}$ and the converse to the dominated convergence theorem. Therefore, $\frac{1}{\varepsilon_k^2} (u_{\varepsilon_k}^-)^2 \rightarrow \infty$ a.e. on A and we find, by Fatou's Lemma, that

$$\liminf_{\ell} \int_A \frac{1}{\varepsilon_{\ell}^2} |u_{\varepsilon_{\ell}}^-|^2 = \infty.$$

This is a contradiction, since $\|\frac{1}{\varepsilon} u_{\varepsilon}^-\|_{L^2} \leq C \|F\|_{(L^2)^*}$, for each $\varepsilon > 0$, by (4.11).

Next, we prove that

$$a(\tilde{u}, v - \tilde{u}) \geq F(\tilde{u}), \forall v \in \dot{H}^1, v \geq 0. \quad (4.29)$$

Indeed, let u_{ε} be the solution of (4.7). By Lemma 4.8, we have

$$\langle G_{\varepsilon}(v), v - u_{\varepsilon} \rangle \geq \langle G_{\varepsilon}(u_{\varepsilon}), v - u_{\varepsilon} \rangle = 0, \forall v \in H^1. \quad (4.30)$$

Since $v^- = 0$ when $v \geq 0$, (4.30) implies

$$a(v, v - u_{\varepsilon}) + \varepsilon \int_{\mathbb{R}^N} (Tv)(v - u_{\varepsilon}) \geq F(v - u_{\varepsilon}), \forall v \in H^1, v \geq 0. \quad (4.31)$$

For each $\varphi \in C_c^{\infty}$, since $\mathcal{F}(\varphi) \in \mathcal{S}$ and $\xi \mapsto \frac{1}{|\xi|^2}$ is integrable near the origin, we have

$\xi \mapsto \mathcal{F}(\varphi)(\xi)/|\xi| \in L^2$, and thus

$$\begin{aligned}
\varepsilon \left| \int_{\mathbb{R}^N} T(\varphi)(\varphi - u_\varepsilon) \right| &= \frac{\varepsilon}{(2\pi)^N} \left| \int_{\mathbb{R}^N} \frac{1}{|\xi| m(\xi/|\xi|)} \mathcal{F}(\varphi)(\xi) \overline{|\xi| \mathcal{F}(\varphi - u_\varepsilon)(\xi)} \right| \\
&\leq C\varepsilon \left(\int_{\mathbb{R}^N} \left(\frac{\mathcal{F}(\varphi)(\xi)}{|\xi| m(\xi/|\xi|)} \right)^2 d\xi \right)^{1/2} \|\nabla(\varphi - u_\varepsilon)\|_{L^2} \\
&\leq C\varepsilon \left(\int_{\mathbb{R}^N} \left(\frac{\mathcal{F}(\varphi)(\xi)}{|\xi|} \right)^2 d\xi \right)^{1/2} \|\nabla(\varphi - u_\varepsilon)\|_{L^2} \\
&\rightarrow 0,
\end{aligned} \tag{4.32}$$

using (HFW) for the last inequality and Lemma 4.6 for the limit.

Passing to the limits $\varepsilon \rightarrow 0$ in (4.31), using (4.32) and the weak continuity in $\dot{H}^1$ of $a(v, \cdot)$ and F , we have

$$a(v, v - \tilde{u}) \geq F(v - \tilde{u}), \forall v \in C_c^\infty, v \geq 0. \tag{4.33}$$

Using the fact that a nonnegative map $v \in \dot{H}^1$ can be approximated with nonnegative maps in C_c^∞ , we have

$$a(v, v - \tilde{u}) \geq F(v - \tilde{u}), \forall v \in \dot{H}^1, v \geq 0. \tag{4.34}$$

Minty's Lemma [12, Lemma 1.5] yields the desired conclusion (4.29) since the operator

$$\dot{H}^1 \ni v \mapsto G(v) \in (\dot{H}^1)^*,$$

with

$$\langle G(v), \varphi \rangle := a(v, \varphi) - F(\varphi), \forall \varphi \in \dot{H}^1,$$

is monotone (this can be shown as in Lemma 4.8). For the convenience of the reader, we reproduce the proof in [12, Lemma 1.5].

Let $v \in \dot{H}^1$ be nonnegative and set $w_t = (1 - t)\tilde{u} + tv \geq 0$ for each $0 \leq t \leq 1$. We have

$$\begin{aligned}
t(a(w_t, v - \tilde{u}) - F(v - \tilde{u})) &= a(w_t, t(v - \tilde{u})) - F(t(v - \tilde{u})) \\
&= a(w_t, w_t - \tilde{u}) - F(w_t - \tilde{u}) \\
&\geq 0,
\end{aligned} \tag{4.35}$$

using (4.34) for the inequality. This implies that

$$a((1 - t)\tilde{u} + tv, v - \tilde{u}) - F(v - \tilde{u}) \geq 0, \forall 0 < t < 1. \tag{4.36}$$

Letting $t \rightarrow 0$, we obtain

$$a(\tilde{u}, v - \tilde{u}) \geq F(v - \tilde{u}), \forall v \in \dot{H}^1, v \geq 0. \quad \square$$

Conclusions and further perspectives. We have shown that minimizers of $\mathcal{E}$ have L^2 density under rather general assumptions. This raises a few natural questions. First, it would be interesting to know whether $\mu \in L^p$ when V is in $W_{\text{loc}}^{2,p}$, $p \neq 2$, instead of H_{loc}^2 . We may also wonder whether the approach presented in our article could be applied to the fractional and 2D counterparts of $\mathcal{E}$. More precisely, replacing W with

$$W_s(x) = \frac{\Psi(x/|x|)}{|x|^{N-2s}}, \quad 0 < s < 1,$$

or with

$$W_{2D}(x) = -\log(|x|) + \Psi(x/|x|),$$

when $N = 2$, we may aim at regularity results on the minimizers of the corresponding energies, under assumptions similar to (HV), (HW), and (HFW). In a forthcoming note [3], we show that our method can be adapted to the latter case.

A Appendix. Existence and uniqueness of a minimizer

This appendix is devoted to the proof of the following result, which holds when (HFW) is relaxed to

$$\mathcal{F}(W) \geq 0. \tag{HFWR}$$

Proposition A.1. *Assume that (HW) and (HFWR) hold and that V is a non-negative, coercive, and lower semi-continuous.*

Then, there exists a unique minimizer of the functional $\mathcal{E}$. Moreover, this minimizer is compactly supported.

We first present a proof of the following fact, arguing as in Mora [17, Proposition 3.2].

Lemma A.2. *Let ν be a positive, finite, and compactly supported measure such that $\mathcal{E}(\nu) < \infty$. We have*

$$\int_{\mathbb{R}^N} W * \nu \, d\nu = \int_{\mathbb{R}^N} \mathcal{F}(W)(\xi) |\mathcal{F}(\nu)|^2 \, d\xi.$$

Proof. We approximate ν with the convolutions $\nu_\varepsilon := \nu * \rho_\varepsilon$, where $\rho \in C_c^\infty$ is a radial mollifier.

Since ν is a finite and compactly supported, ν_ε is in C_c^∞ and $\mathcal{F}(W * \nu_\varepsilon) = \mathcal{F}(W) \mathcal{F}(\nu) \mathcal{F}(\rho_\varepsilon)$. For each $\varepsilon > 0$, the Plancherel formula yields

$$\int_{\mathbb{R}^N} (W * \nu_\varepsilon) \, d\nu_\varepsilon = \int_{\mathbb{R}^N} \mathcal{F}(W)(\xi) |\mathcal{F}(\nu)(\xi)|^2 |\mathcal{F}(\rho_\varepsilon)(\xi)|^2 \, d\xi. \tag{A.1}$$

We first study the convergence of the left-hand side in (A.1). Since

$$\int_{\mathbb{R}^N} (W * v_\varepsilon)(x) v_\varepsilon(x) dx = \left((W * v_\varepsilon) * \check{v}_\varepsilon \right) (0) = \left(W * v * \rho_\varepsilon * \check{v} * \rho_\varepsilon \right) (0),$$

where $\check{v}(x) := v(-x)$, we have, by Tonelli's theorem,

$$\begin{aligned} \int_{\mathbb{R}^N} (W * v_\varepsilon)(x) v_\varepsilon(x) dx &= \left(W * (\rho_\varepsilon * \rho_\varepsilon) * v * \check{v} \right) (0) \\ &= \int_{\mathbb{R}^N} \int_{\mathbb{R}^N} (W * \eta_\varepsilon)(x - y) dv(x) dv(y), \end{aligned} \tag{A.2}$$

where $\eta_\varepsilon = \rho_\varepsilon * \rho_\varepsilon$ is also a radial mollifier with compact support. Since W is continuous outside the origin and $\lim_{x \rightarrow 0} W(x) = \infty$, we have that $W * \eta_\varepsilon(x) \rightarrow W(x)$ for every $x \in \mathbb{R}^N$. On the other hand, we have, for each $x \in \mathbb{R}^N$,

$$(W * \eta_\varepsilon)(x) \leq C(E * \eta_\varepsilon)(x) \leq CE(x) \leq CW(x).$$

Here, we use (HW) for the first inequality and for the last one, while the second inequality is a consequence of the superharmonicity of E (Lemma 2.4), since η is a radial mollifier. Since $\mathcal{E}(v) < \infty$, we have

$$\int_{\mathbb{R}^N} (W * v_\varepsilon)(x) v_\varepsilon(x) dx = \int_{\mathbb{R}^N} (W * \eta_\varepsilon)(x - y) dv(x) dv(y) \rightarrow \int_{\mathbb{R}^N} (W * v)(x) dv(x),$$

by dominated convergence.

We next study the convergence of the right-hand side in (A.1). We have $\mathcal{F}(\rho_\varepsilon)(\xi) \rightarrow 1$, for each $\xi \in \mathbb{R}^N$. Thanks to the positivity of $\mathcal{F}(W)$, we therefore find that

$$\mathcal{E}(v) = \liminf \int_{\mathbb{R}^N} W * v_\varepsilon dv_\varepsilon \geq \int_{\mathbb{R}^N} \mathcal{F}(W)(\xi) |\mathcal{F}(v)(\xi)|^2 d\xi,$$

by Fatou's lemma. Thus, $\mathcal{F}(W) |\mathcal{F}(v)|^2 \in L^1$ and since $\|\mathcal{F}(\rho_\varepsilon)\|_{L^\infty}$ is uniformly bounded, this allows us to obtain

$$\int_{\mathbb{R}^N} \mathcal{F}(W)(\xi) |\mathcal{F}(v)(\xi)|^2 |\mathcal{F}(\rho_\varepsilon)(\xi)|^2 d\xi \rightarrow \int_{\mathbb{R}^N} \mathcal{F}(W)(\xi) |\mathcal{F}(v)(\xi)|^2 d\xi,$$

by dominated convergence. □

Before turning to the proof of Proposition A.1, we state a last useful lemma.

Lemma A.3. *Let f and g be real analytic functions defined on $\mathbb{R}^N$ such that $f = g$ on a set A of positive Lebesgue measure. Then, we have $f = g$.*

The proof Proposition A.1 we present below essentially follows the arguments used in [18, Section 2], [16, Proposition 3.1], and [19, Lemma 2.10, Theorem 2.1].

Proof of Proposition A.1. Existence of a minimizer. This fact follows from the direct method. Let

(μ_n) be a minimizing sequence of $\mathcal{E}$ satisfying, with no loss of generality, $\sup_k \mathcal{E}(\mu_k) < \infty$. For each $\varepsilon > 0$, we find $R > 0$ such that $V(x) \geq 1/\varepsilon$ in B_R^c , by coercivity of V . We thus have, for each n ,

$$\int_{B_R^c} 1/\varepsilon \, d\mu \leq \int_{B_R^c} V \, d\mu \leq \mathcal{E}(\mu_n) \leq \sup_k \mathcal{E}(\mu_k),$$

using the positivity of V and W for the second inequality. This reads

$$\mu_n(B_R^c) \leq \varepsilon \sup_k \mathcal{E}(\mu_k), \forall n,$$

meaning that the sequence (μ_n) is tight. By Prokhorov's theorem, we may therefore find $\mu \in \mathcal{P}(\mathbb{R}^N)$ such that, up to extraction, $\mu_n \rightarrow \mu$ in the sense of probability measures. Since $\mathcal{E}$ is l.s.c. for the weak convergence of probability measures (see, e.g., [13, Lemma 0.1 and (1.4.4)]), we find that

$$\mathcal{E}(\mu) \leq \liminf_k \mathcal{E}(\mu_k) = \inf \mathcal{E},$$

and thus μ is a minimizer of $\mathcal{E}$.

Minimizers of $\mathcal{E}$ are compactly supported. This is obtained by repeating the arguments in Serfaty [19, Proof of Theorem 2.1, Step 1].

Uniqueness. By the above, it suffices to prove that $\mathcal{E}$ is strictly convex on the space of compactly supported probability measures ν such that $\mathcal{E}(\nu) < \infty$. Let ν_1, ν_2 be such probability measures. Setting $\nu_t = (1-t)\nu_1 + t\nu_2$, for each $0 < t < 1$, we have

$$\int_{\mathbb{R}^N} (W * \nu_t) \, d\nu_t = \int_{\mathbb{R}^N} \mathcal{F}(W)(\xi) |(1-t)\mathcal{F}(\nu_1)(\xi) + t\mathcal{F}(\nu_2)(\xi)|^2 \, d\xi,$$

by Lemma A.2. But $\mathbb{C} \ni z \mapsto |z|^2$ is convex, thus we have

$$\begin{aligned} \int_{\mathbb{R}^N} (W * \nu_t) \, d\nu_t &= \int_{\mathbb{R}^N} \mathcal{F}(W)(\xi) |(1-t)\mathcal{F}(\nu_1)(\xi) + t\mathcal{F}(\nu_2)(\xi)|^2 \, d\xi \\ &\leq (1-t) \int_{\mathbb{R}^N} \mathcal{F}(W)(\xi) |\mathcal{F}(\nu_1)(\xi)|^2 \, d\xi + t \int_{\mathbb{R}^N} \mathcal{F}(W)(\xi) |\mathcal{F}(\nu_2)(\xi)|^2 \, d\xi \\ &= (1-t) \int_{\mathbb{R}^N} W * \nu_1 \, d\nu_1 + t \int_{\mathbb{R}^N} W * \nu_2 \, d\nu_2, \end{aligned} \tag{A.3}$$

using (HFWR).

We now show that if, for some $t \in (0, 1)$, we have equality in (A.3), then $\nu_1 = \nu_2$. Indeed, consider a set $A \subset \mathbb{R}^N$ of positive Lebesgue measure where $\mathcal{F}(W) > 0$. If there is equality in (A.3), then $\mathcal{F}(\nu_1) = \mathcal{F}(\nu_2)$ on A , since $z \mapsto |z|^2$ is strictly convex. But $\mathcal{F}(\nu_{1,2})$ are real analytic functions, since $\nu_{1,2}$ are compactly supported (this is an easy case of the Paley–Wiener–Schwartz theorem). Therefore, Lemma A.3 implies that $\mathcal{F}(\nu_1) = \mathcal{F}(\nu_2)$ everywhere and this yields the desired conclusion. $\square$

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